Gustavo Diaz, Northwestern Political Science, 2001 Sheridan Rd, Suite 3110, Evanston IL 60208

Assistant Professor Search, Data Science

Dear Members of the Search Committee,

I write to express my interest in your Assistant Professor search in data science. I am an Assistant Professor of Instruction in the Department of Political Science at Northwestern University, where I teach courses on statistics, statistical computing, and computational social science and conduct research in political methodology and applied statistics. My work is published in leading social science outlets including *Political Analysis*, the *British Journal of Political Science*, *World Development*, and the *Journal of Experimental Political Science*.

My research agenda focuses on using tools from statistics and data science to improve research design before data collection. My current focus is on statistical precision. This is overlooked in the statistics, econometrics, and political methodology literature in favor of identifying unbiased estimators. Implicitly, this literature assumes that one can improve statistical precision by just increasing sample size. This is not feasible in many social science applications due to practical or ethical considerations.

Focusing on surveys and experiments, this agenda seeks to shape applied research by focusing on cases where one can seemingly improve statistical precision without sacrificing unbiasedness. As I show in my work, this usually implies unforeseen costs in other dimensions.

For example, in "Balancing Precision and Retention in Experimental Design", published in *Political Analysis*, I discuss how implementing alternatives to the standard experimental design, such as block randomization or repeated measures, may attenuate expected precision gains via explicit or implicit sample loss, a concern that prevents researchers from applying these techniques widely. Through three replications and six reanalyses of previously published experiments in leading political science journals, I show how precision gains from alternative designs can withstand significant degrees of sample loss.

As another example, in a solo-authored publication in the *Journal of Experimental Political Science*, I discuss the unforeseen costs of implementing double list experiments. This is a variant of the list experiment that promises narrower confidence intervals but comes with under-explored questionnaire design complications in the form of carryover design effects, a special kind of question order effect. I introduce statistical tests to diagnose this effect, which in turn facilitate the implementation of a more efficient technique.

My work also produces insights for substantive research in in political science. For example, I am part of an *European Research Council Starting Grant* (~1.5 million USD) on understanding criminal governance in unexpected contexts. Many studies of criminal governance in Latin America focus on countries with powerful criminal organizations and high levels of state coercive capacity, we focus instead on countries where criminal groups are less powerful yet entrenched, despite state presence being extended beyond security

policies. Our multi-year grant conducts comparative public opinion surveys in Argentina, Chile, Costa Rica, and Uruguay. Our first product from this grant just received a **revise and resubmit** decision in *Latin American Politics and Society*. An important component of our grant is to advance innovation in survey research methods. I outline these developments in my research statement.

As another example, in an article published in the *British Journal of Political Science*, I use an information experiment in Argentina to study gendered differential reactions to policy implementation. Previous work suggests that women face higher scrutiny for their performance in office. However, in the context of the implementation of a food distribution program, I find that voters are only responsive to performance information among men officeholders and tend to ignore performance information when told that an officeholder is a woman. This stems from voters' perception of men politicians as the default category, and the tendency to believe that women politicians' good performance stems from factors beyond the incumbent's control. This finding calls attention to an unforeseen informational imbalance on survey experimental work in the subject, which future studies should take into account.

My teaching focuses on making quantitative methods accessible to diverse audiences. I accomplish this through a combination of flexibility and accountability. At Northwestern, I am the central person teaching methods courses in the department. I teach the first course in the PhD sequence on Probability and Statistics, a PhD elective on Machine Learning, the required undergrad-level Introduction to Empirical Methods, and an undergraduate research seminar on how machine learning and experimentation are used in government, industry, and civil society to inform decision-making. In the Spring, I will add a hands-on undergraduate course on survey research methods to my portfolio.

I believe my experience teaching statistics courses in a PhD program is commesurate to that required to offer courses at all levels in an undergraduate program in data science. Beyond the classroom, I routinely work with and advise undergraduates on topics related to applied statistics. Last year, I served as a mentor for the Farrell Fellowship program at Northwestern, which paired me two student mentees to collaborate on various projects on experimental design. I also routinely advice senior theses across the social sciences, including Northwestern's program on Mathematical Methods for the Social Sciences.

I am prepared to teach courses on statistical inference, causal inference, machine learning, computational social science, statistical computing, and data-driven decision-making. You can find copies of current and future syllabi in my teaching portfolio.

I believe my expertise makes me an excellent fit at Lewis & Clark. If you have any questions, you can contact me via email or phone.

Sincerely,

Gustavo Diaz
Northwestern University